

Ipe Showcase & Best Practices

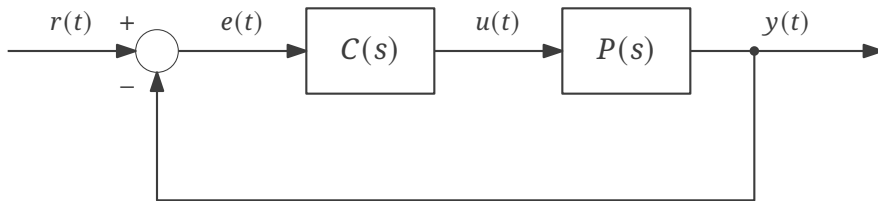
Comprehensive Demonstration

- **1. Geometric Drawings & Block Diagrams:** CAD-like snapping
- **2. Stepwise Plot Animations:** Layer/View hierarchy with legend-by-legend curve reveals
- **3. In-Slide Video Presentation:** Seamless playback with Pympress integration
- **4. LaTeX Typography & Tables:** Academic tables and mathematical formulas

Geometric Drawings

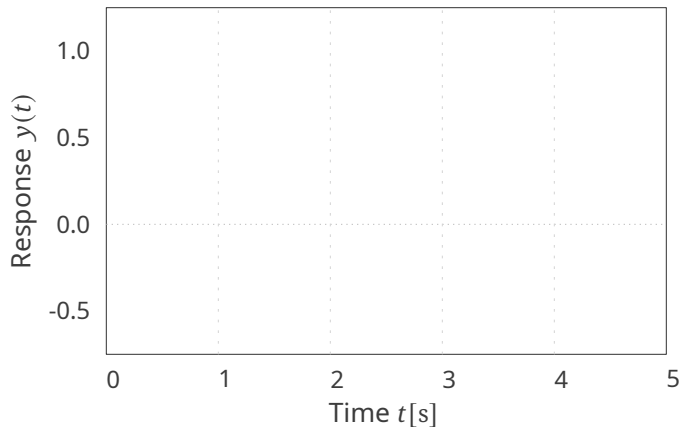
Block Diagram with Snapping & LaTeX Math

- CAD-like exact snapping
- $\LaTeX$ math inside blocks



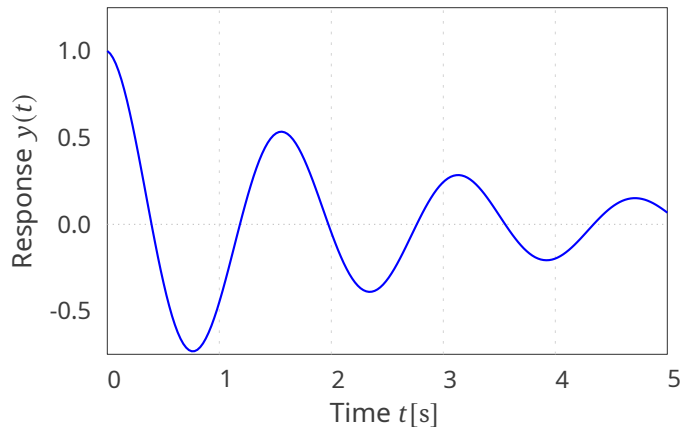
Stepwise Plot Animations

Step Response: Parameter Variation (ζ)



Stepwise Plot Animations

Step Response: Parameter Variation (ζ)

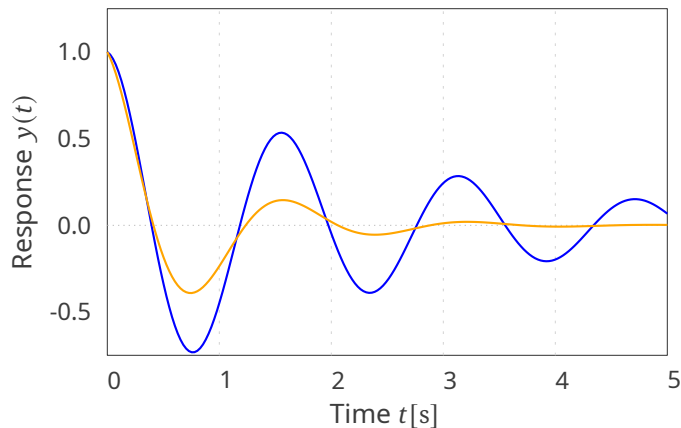


— $\zeta = 0.1$ (Underdamped)

Large overshoot with sustained oscillation

Stepwise Plot Animations

Step Response: Parameter Variation (ζ)



— $\zeta = 0.1$ (Underdamped)

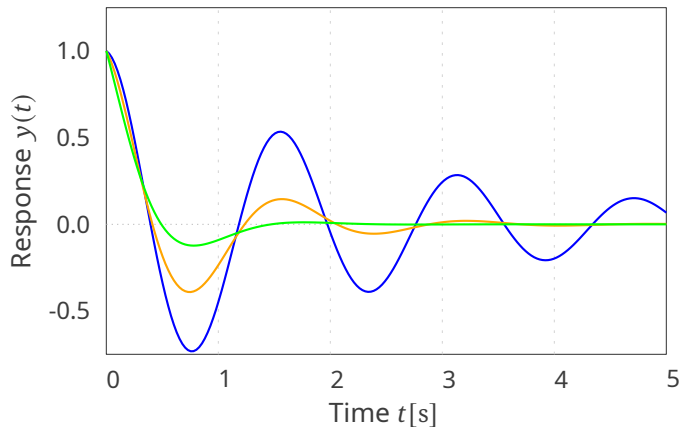
Large overshoot with sustained oscillation

— $\zeta = 0.3$ (Moderate)

Faster convergence with reduced peak

Stepwise Plot Animations

Step Response: Parameter Variation (ζ)



— $\zeta = 0.1$ (Underdamped)

Large overshoot with sustained oscillation

— $\zeta = 0.3$ (Moderate)

Faster convergence with reduced peak

— $\zeta = 0.6$ (Well-damped)

Minimal overshoot, fast settling time

Embedded Video Presentation

In-Slide Video Playback (Pympress)

- **Direct Video in PDF**

- WebM (VP8/VP9) / GIF / MP4
- Native GTK overlay in Pympress
- Auto-loop & control bar

- **Workflow**

- Ipelets > Media
- Insert video & export PDF
- Run: `pympress presentation.pdf`



Academic Three-Line Table & Math Formulas

Parameter	Symbol	Nominal Value	Range	Unit
Natural Frequency	ω_n	4.0	1.0 – 10.0	rad/s
Damping Ratio	ζ	0.3	0.05 – 1.0	–
Sampling Interval	T_s	1.0	0.1 – 5.0	ms
Steady-State Gain	K_p	1.0	0.5 – 2.0	V/V

- Professional publication-ready table formatting using standard $\text{\LaTeX}$ syntax
- Consistent mathematical typography matching the parent journal / conference paper

Appendix: Next Steps

- **Guide:** See `docs/GUIDE.md` for walkthrough
- **Templates:** Copy clean starting files from `templates/`
- **Plot Importer:** Use Ipelets > Insert Vector Plot for MATLAB / Python figures
- **Handout Export:** Use Ipelets > Export Handout (Exclude Appendix) for clean printouts

